n

* Linear Regression ⇒ Probabilistic View

random → $y^{(i)} = \underbrace{\theta^T x^{(i)}}_{\text{deterministic}} + \epsilon^{(i)}$ → random

↳ a noise term brings expected y value closer to real value.

where, $\epsilon^{(i)} \sim N(0, \sigma^2)$ → mean

↳ normal distribution

$\theta \in \mathbb{R}^{d+1}$

eg ⇒ $y = 15$

$\hat{y} = 14.8$

ϵ takes care of rest 0.2

σ^2 ⇒ variance

↳ std. dev.

PDF Probability Density Function for normal dist. ↑

$$P(\epsilon^{(i)}) \Rightarrow \frac{1}{\sigma\sqrt{2\pi}} \exp\left[\frac{-1}{2} \frac{(\epsilon^{(i)} - 0)^2}{\sigma^2}\right]$$

exp ↓ e

ϵ this is gaussian noise for linear ~~regres~~ regression

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$$P(\epsilon^{(i)}) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[\frac{-1}{2}\frac{(\epsilon^{(i)}-0)^2}{\sigma^2}\right]$$

$$\underbrace{y^{(i)}}_{\theta} = \theta^T x^{(i)} + \epsilon^{(i)}$$

$$\epsilon^{(i)} = y^{(i)} - \theta^T x^{(i)}$$

$$\epsilon^{(i)} \sim N(0,\sigma^2) \underset{\text{implies}}{\Rightarrow} \overbrace{y^{(i)}}^{\text{random} \to \text{varies}} - \underbrace{\theta^T x^{(i)}}_{\text{constant}} \sim N(0,\sigma^2)$$

Gaussian's have location scale property

$$y^{(i)} \sim N(\underbrace{\theta^T x^{(i)}}, \sigma^2)$$

↳ The mean has changed to a constant

↳ mean is diff. for every sample

$$\underset{\text{conditional}}{P(y^{(i)}} \mid x^{(i)} \underset{\text{parameterized by } \theta}{;\ \theta)} = \cancel{\frac{1}{\sigma\sqrt{2\pi}} \exp\left[\frac{-1}{2}(\epsilon^{(i)} - 0\right.}$$

$$P(y^{(i)} \mid x^{(i)};\theta) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[\frac{-1}{2}\left\{\frac{(y^{(i)} - \overbrace{\theta^T x^{(i)}}^{\text{mean}})^2}{\sigma^2}\right.\right]$$

$$\underset{\text{all the samples}}{P(y \mid x} ; \theta) = \underbrace{\prod_{i=1}^{n}}_{\text{b/c we assume [IID]}} P(y^{(i)} \mid \underset{\text{one sample "i"}}{x^{(i)}})$$

eq.

Step 1 ⇒ MLE estimation

$$L(\theta; x, y) = P(y \mid x; \theta)$$

$$L(\theta; x, y) = p(y \mid x; \theta)$$ from eq.

$$= \prod_{i=1}^{n} p(y^{(i)} \mid x^{(i)})$$

$$L(\theta; x, y) \Rightarrow \prod_{i=1}^{n} \frac{1}{\sigma\sqrt{2\pi}} \exp\left[\frac{-1}{2} \frac{\left(y^{(i)} - \theta^T x^{(i)}\right)^2}{\sigma^2}\right]$$

Step 2:- Take ~~logrithm~~ logarithm of likelihood function

$$l(\theta) = \log\left(\prod_{i=1}^{n} \frac{1}{\sigma\sqrt{2\pi}} \exp\left[\frac{-1}{2} \frac{\left(y^{(i)} - \theta^T x^{(i)}\right)^2}{\sigma^2}\right]\right)$$

$$\Rightarrow \sum_{i=1}^{n} \log \frac{1}{\sigma\sqrt{2\pi}} + \log\left(\exp\left[\frac{-1}{2} \frac{\left(y^{(i)} - \theta^T x^{(i)}\right)^2}{2}\right]\right)$$

$\prod \rightarrow \Sigma$ bcoz multiply becomes addition under log

$$\Rightarrow \underbrace{n \log \frac{1}{\sigma\sqrt{2\pi}}}_{\because \frac{1}{\sigma\sqrt{2\pi}} \text{ are constants}} + \underbrace{\sum_{i=1}^{n} -\frac{1}{2}\left[\frac{\left(y^{(i)} - \theta^T x^{(i)}\right)^2}{\sigma^2}\right]}_{\text{log and exp (e) get cancelled}}$$

$$\boxed{l(\theta) \Rightarrow n \log \frac{1}{\sigma\sqrt{2\pi}} - \cancel{\frac{1}{2\sigma^2}} \frac{1}{2\sigma^2} \sum_{i=1}^{n} \underbrace{\left(y^{(i)} - \theta^T x^{(i)}\right)^2}}$$

Cost function

maximizes $l(\theta)$ if we minimize this bracket ~~te~~ term

$$l(\theta)\uparrow \Rightarrow \left(y^{(i)} - \theta^T x^{(i)}\right)^2 \downarrow$$

$$\underset{\theta}{\operatorname{argmax}}\ l\theta \Rightarrow \underset{\theta}{\operatorname{argmin}}\ J(\theta)$$

Finally find the theta "θ"

$10 \times \frac{10}{100} \to 0.1 \times 10 \to \underline{1}$

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- Find θ, by executing gradient descent or normal eqns approach on $\boxed{\underset{\theta}{\operatorname{argmin}}\ J(\theta)}$

~~Classification dataset~~ → ~~dataset~~

* Logisti

Datu

binary →